UPI TRANSACTIONS DATA ANALYSIS

Project Report | Tableau Dashboard & Excel Dataset

20,000 Transactions	**80%** Successful	**20%** Failed	**4** Cities

Prepared from the supplied workbook: UPI+Transactions.xlsx and the supplied Tableau dashboard PDF. The report distinguishes dataset calculations from dashboard observations and flags issues that should be corrected before presenting the dashboard as a reliable business report.

Project focus: transaction status, city, payment method, merchant, customer age, bank, device, payment mode and transaction purpose.

1. Executive Summary

This project analyzes 20,000 transaction records dated 01 Jan 2024 to 30 Dec 2024. The dataset contains transaction amount, transaction status, city, bank details, payment method, merchant, purpose, device, payment mode, customer age and currency. The analysis finds 16,000 successful transactions (80.0%) and 4,000 failed transactions (20.0%).

The average recorded transaction amount is 993.61 and the median is 983.39. These are descriptive values in the source data, but they must not be interpreted as one unified currency amount because the workbook mixes USD, EUR, GBP and INR.

Overall conclusion: The dashboard provides a useful starting point for exploring transaction dimensions, but several data and labeling issues need correction. In particular, the dashboard uses Amount aggregations in charts whose titles say “Total Transactions”, and the dataset contains strong patterned relationships that may not represent real-world payment behavior.

2. Project Objectives

• Measure successful and failed transaction counts.
• Compare transaction activity across cities, merchants, payment methods and banks.
• Explore customer age groups, devices, payment modes and transaction purposes.
• Build an interactive Tableau view with filters and geographic comparison.
• Identify data-quality limitations and recommend improvements before business use.

3. Dataset Overview

The workbook contains one sheet (Sheet1), 20,000 rows and 19 columns. The observed date range is 1 January 2024 to 30 December 2024. All 19 columns have complete values in the supplied file; no fully duplicated rows or duplicate TransactionID values were found.

Field	Description
TransactionID	Unique transaction identifier
TransactionDate	Date of transaction
Amount	Recorded transaction amount
BankNameSent / BankNameReceived	Sending and receiving bank labels
RemainingBalance	Recorded balance after transaction
City, Gender, CustomerAge	Customer and location attributes
TransactionType, Status	Payment/transfer type and success/failure status
DeviceType, PaymentMethod, PaymentMode	Device, initiation method and timing mode
MerchantName, Purpose	Merchant and transaction purpose
Currency	Currency code attached to the record
CustomerAccountNumber / MerchantAccountNumber	Account-number fields in the dataset

4. Methodology

The analysis was completed by loading the Excel workbook, checking row and column counts, validating dates, checking missing values and duplicates, and summarizing records with counts, averages and grouped comparisons. Age bands in this report are calculated as 20–29, 30–39, 40–49 and 50–59. The supplied dashboard PDF was reviewed separately to identify its visible charts, filters and displayed aggregations.

Suggested Tableau workflow: connect to the Excel file; verify data types; create calculated fields for transaction count and success rate; build worksheets for status, city, merchant, payment method and age; add filters; then assemble the worksheets into a dashboard.

5. Key Findings from the Excel Dataset

Metric	Result
Total records	20,000
Successful transactions	16,000 (80.0%)
Failed transactions	4,000 (20.0%)
Average amount	993.61 (source currency varies)
Median amount	983.39 (source currency varies)
Date range	01 Jan 2024 – 30 Dec 2024
Missing cells / duplicate rows	0 / 0

5.1 City distribution

City	Transactions	Amount sum*	Currency
Bangalore	5,000	4,875,336.23	EUR
Delhi	5,000	5,036,738.71	USD
Hyderabad	5,000	4,907,634.17	GBP
Mumbai	5,000	5,052,564.92	INR

*The amount sums are shown as stored and are not a cross-currency total. Each city is associated with only one currency in this dataset: Delhi–USD, Bangalore–EUR, Hyderabad–GBP and Mumbai–INR.

Each city contains exactly 5,000 records. Mumbai has the largest recorded amount sum (5,052,564.92 INR) and Bangalore the smallest (4,875,336.23 EUR), but these totals are not directly comparable because they use different currencies.

5.2 Payment method and device

Payment method	Transactions	Amount sum*
Phone Number	6,667	6,640,042.16
QR Code	6,667	6,589,344.39
UPI ID	6,666	6,642,887.48

*The workbook links payment method and device categories in a highly regular pattern; confirm the source data before interpreting this as user preference.

5.3 Merchant and status

Merchant	Transactions	Success	Failed
Amazon	4,000	4,000	0
Flipkart	4,000	0	4,000
IRCTC	4,000	4,000	0
Swiggy	4,000	4,000	0
Zomato	4,000	4,000	0

Important pattern: all 4,000 Flipkart records are marked Failed, while all records for Amazon, IRCTC, Swiggy and Zomato are marked Success. This perfectly separated outcome pattern is unusual and should be checked against the original source or generation logic; it should not be presented as a real merchant performance conclusion without validation.

5.4 Age and monthly activity

Age band	Transactions
20–29	5,000
30–39	5,000
40–49	5,000
50–59	5,000

Month	Transactions
Jan	1,666
Feb	1,667
Mar	1,667
Apr	1,667
May	1,667
Jun	1,667
Jul	1,667
Aug	1,667
Sep	1,667
Oct	1,666
Nov	1,666
Dec	1,666

Each age band contains exactly 5,000 records. Monthly counts are also almost perfectly even (1,666 or 1,667 per month). These balanced distributions may reflect a constructed or sample dataset rather than naturally observed transaction activity.

6. Tableau Dashboard Review

The supplied four-page PDF shows a dashboard titled “Tableau Dashboard 2”. The main dashboard includes filters for receiving bank, currency, device type, payment mode and purpose. Visible views include an age-group summary, a city map and a grouped bar chart by payment method and merchant. A second page shows the payment-method/merchant chart with filter controls; later pages show the age-group view and city map separately.

What works well

• Filters support interactive exploration by currency, device, payment mode, purpose and receiving bank.
• The city map gives a geographic overview of the four city categories.
• The merchant/payment-method chart allows side-by-side comparisons.
• The dashboard combines geographic and categorical views in one report.

Corrections recommended before publishing

1. **Fix chart titles and measures.** The city map labels (about 4.88M–5.05M) match SUM(Amount) by city, not transaction counts. The payment-method/merchant bars also show SUM(Amount). Rename them to “Recorded Amount by City” and “Recorded Amount by Payment Method & Merchant”, or change the measure to COUNT(TransactionID) if the intention is transaction volume.
2. **Handle currency correctly.** The workbook contains four currencies, and each city maps to a single currency. Do not sum USD, EUR, GBP and INR into one KPI. Use a currency filter, separate currency panels, or convert amounts using a documented exchange-rate date and source.
3. **Verify the age-group visual.** The PDF shows 3,000, 5,000 and 12,000 across three groups, while the workbook has 5,000 records in each of the four standard age bands (20–29, 30–39, 40–49, 50–59). Rebuild this view from CustomerAge and validate the grouping logic.
4. **Investigate merchant/status pattern.** All failures are associated with Flipkart. Confirm whether this is intentional sample data or a data-generation issue.
5. **Improve naming.** Replace the generic dashboard title with “UPI Transactions Analysis Dashboard” and label KPIs with units and definitions.

7. Data Quality, Assumptions and Limitations

• No blank cells, fully duplicated rows or duplicate transaction IDs were detected in the workbook.
• The dataset includes USD, EUR, GBP and INR. An overall sum or average across these currencies is not financially meaningful without currency conversion.
• City and currency are perfectly linked in the supplied data, which prevents independent comparison of city and currency effects.
• Merchant and status are strongly linked: every Flipkart record is failed and every other merchant record is successful. This may be a synthetic-data artifact.
• Categories and monthly/age distributions are unusually balanced. Results should be treated as descriptive of this file only, not as representative of the UPI ecosystem.
• Account-number fields are included in the source workbook. They are not required for the dashboard and should be excluded or masked in any public-facing report.

8. Recommendations and Future Scope

• Validate the source dataset and confirm whether it is synthetic, simulated or production data.
• Add KPI cards for total transaction count, successful count, failed count and success rate.
• Add a monthly trend for transaction count and success rate, not only amount.
• Compare success/failure by payment method, device, bank and purpose after validating the patterns.
• Add drill-down filters and informative tooltips with clear metric definitions.
• If real multi-currency analysis is required, normalize to a chosen reporting currency using documented exchange rates.
• Remove account identifiers from published extracts and follow data privacy practices.

9. Conclusion

The project demonstrates an end-to-end analytical workflow using an Excel transaction dataset and a Tableau dashboard. The workbook contains 20,000 records across 2024, with an 80% success rate and 20% failure rate. City, payment method, merchant, age and status views offer useful ways to explore the data. However, currency mixing, mislabeled amount aggregations and highly regular category relationships must be addressed before drawing business conclusions. With these corrections, the dashboard can become a clearer and more reliable portfolio project.

Source files: UPI+Transactions.xlsx (Sheet1) and Dashboard.pdf (4-page Tableau dashboard export). All numerical findings in Sections 1, 3 and 5 were calculated from the supplied workbook; dashboard observations are based on the supplied PDF.